



Python

Python SDK Usage

Copy page



We provide a [PyPI package](#), `vastai-sdk`, for convenient Python usage.

PyPI Install

You can install the latest stable PyPI release with:

Text

```
pip install vastai-sdk
```

Usage

Import the package:

Text

```
from vastai_sdk import VastAI
```

Construct a Vast client with your API key:

Text

```
vast_sdk = VastAI(api_key='YOUR_API_KEY')
```

Resource Methods



>

For example, the CLI command `vastai show instances` has the equivalent,
`vast_sdk.show_instances()` .

Text

```
output = vast_sdk.show_instances()  
print(output)
```

Getting help

Use the built-in `help()` method to view detailed documentation for any SDK method. This will show you the method's description, parameters, query syntax, and usage examples.

Text

```
help(vast_sdk.search_offers)
```

Example Usage

Here are some example usages of our Python SDK class `VastAI` :

Search offers

Find an available RTX 3090 GPU

Text

```
vast_sdk.search_offers(query='gpu_name=RTX_5090 rented=False rentable=True')
```

Starting and Stopping Instances



>

Creating a New Instance

Create a new instance based on given parameters (performs search offers + create instance).

Text

```
vast_sdk.launch_instance(num_gpus="1", gpu_name="RTX_3090", image="pytorch/pyt
```

Copying Files Between Instances

Text

```
vast_sdk.copy(src='source_path', dst='destination_path', identity='identity_fi
```

Managing SSH Keys

Create a new SSH key, show all SSH keys, and delete an SSH key.

Text

```
vast_sdk.create_ssh_key(ssh_key='your_ssh_key')

ssh_keys = vast_sdk.show_ssh_keys()
print(ssh_keys)

vast_sdk.delete_ssh_key(ID=123456)
```

Contribution and Issue Reporting



>

Available Methods

Below is a list of the available methods you can call on the **VastAI** client. These methods are categorized for better readability.

Instance Management

Method	Description
<code>start_instance(ID: int)</code>	Start an instance.
<code>stop_instance(ID: int)</code>	Stop an instance.
<code>reboot_instance(ID: int)</code>	Reboot an instance.
<code>destroy_instance(id: int)</code>	Destroy an instance.
<code>destroy_instances(ids: List[int])</code>	Destroy multiple instances.
<code>recycle_instance(ID: int)</code>	Recycle an instance.
<code>label_instance(id: int, label: str)</code>	Label an instance.
<code>show_instance(id: int)</code>	Show details of an instance.
<code>show_instances(quiet: bool = False)</code>	Show all instances.
<code>logs(INSTANCE_ID: int, tail: Optional[str] = None)</code>	Retrieve logs for an instance.
<code>execute(ID: int, COMMAND: str)</code>	Execute a command on an instance.
<code>launch_instance(...)</code>	Launch a new instance with various parameters.

SSH Key Management



>

<code>show_ssh_keys()</code>	Show all SSH keys.
<code>attach_ssh(instance_id: int, ssh_key: str)</code>	Attach an SSH key to an instance.
<code>detach_ssh(instance_id: int, ssh_key_id: str)</code>	Detach an SSH key from an instance.

API Key Management

Method	Description
<code>create_api_key(name: Optional[str] = None, ...)</code>	Create a new API key.
<code>delete_api_key(ID: int)</code>	Delete an API key.
<code>reset_api_key()</code>	Reset the API key.
<code>show_api_key(id: int)</code>	Show details of an API key.
<code>show_api_keys()</code>	Show all API keys.
<code>set_api_key(new_api_key: str)</code>	Set a new API key.

Autoscaler Management

Method	Description
<code>create_autoscaler(test_workers: int = 3, ...)</code>	Create a new autoscaler.
<code>update_autoscaler(ID: int, min_load: Optional[float] = None, ...)</code>	Update an autoscaler.
<code>delete_autoscaler(ID: int)</code>	Delete an autoscaler.
<code>show_autoscalers()</code>	Show all autoscalers.

Endpoint Management



>

<code>delete_endpoint(ID: int)</code>	Delete an endpoint.
<code>show_endpoints()</code>	Show all endpoints.

File Management

Method	Description
<code>copy(src: str, dst: str, identity: Optional[str] = None)</code>	Copy files between instances.
<code>cloud_copy(src: Optional[str] = None, dst: Optional[str] = "/workspace", ...)</code>	Copy files between cloud and instance.
<code>cancel_copy(dst: str)</code>	Cancel a file copy operation.
<code>cancel_sync(dst: str)</code>	Cancel a file sync operation.
<code>scp_url(id: int)</code>	Get the SCP URL for transferring files to/from an instance.

Team Management

Method	Description
<code>create_team(team_name: Optional[str] = None)</code>	Create a new team.
<code>destroy_team()</code>	Destroy a team.
<code>invite_team_member(email: Optional[str] = None, role: Optional[str] = None)</code>	Invite a new member to the team.
<code>remove_team_member(ID: int)</code>	Remove a member from the team.
<code>create_team_role(name: Optional[str] = None, permissions: Optional[str] = None)</code>	Create a new team role.
<code>remove_team_role(NAME: str)</code>	Remove a role from the team.



>

<code>show_team_role(NAME: str)</code>	Show details of a specific team role.
<code>show_team_roles()</code>	Show all team roles.

Host Management

Method	Description
<code>cleanup_machine(ID: int)</code>	Clean up a machine's configuration and resources.
<code>list_machine(ID: int, price_gpu: Optional[float] = None, price_disk: Optional[float] = None, price_inetu: Optional[float] = None, price_inetd: Optional[float] = None, discount_rate: Optional[float] = None, min_chunk: Optional[int] = None, end_date: Optional[str] = None)</code>	List details of a single machine with optional pricing and configuration parameters.
<code>list_machines(IDs: List[int], price_gpu: Optional[float] = None, price_disk: Optional[float] = None, price_inetu: Optional[float] = None, price_inetd: Optional[float] = None, discount_rate: Optional[float] = None, min_chunk: Optional[int] = None, end_date: Optional[str] = None)</code>	List details of multiple machines with optional pricing and configuration parameters.
<code>remove_defjob(id: int)</code>	Remove the default job from a machine.
<code>set_defjob(id: int, price_gpu: Optional[float] = None, price_inetu: Optional[float] = None, price_inetd: Optional[float] = None, image: Optional[str] = None, args: Optional[List[str]] = None)</code>	Set a default job on a machine with specified parameters.
<code>set_min_bid(id: int, price: Optional[float] = None)</code>	Set the minimum bid price for a machine.
<code>schedule_maint(id: int, sdate: Optional[float] = None, duration: Optional[float] = None)</code>	Schedule maintenance for a machine.
<code>cancel_maint(id: int)</code>	Cancel scheduled maintenance for a machine.
<code>unlist_machine(id: int)</code>	Unlist a machine from being available for new jobs.



>

Other Methods

Method	Description
<code>get_gpu_names()</code>	Returns a set of GPU names available on Vast.ai.
<code>show_connections()</code>	Show all connections.
<code>show_deposit(ID: int)</code>	Show deposit details for an instance.
<code>show_earnings(quiet: bool = False, start_date: Optional[str] = None, end_date: Optional[str] = None, machine_id: Optional[int] = None)</code>	Show earnings information.
<code>show_invoices(quiet: bool = False, start_date: Optional[str] = None, end_date: Optional[str] = None, ...)</code>	Show invoice details.
<code>show_ipaddrs()</code>	Show IP addresses.
<code>show_user(quiet: bool = False)</code>	Show user details.
<code>show_subaccounts(quiet: bool = False)</code>	Show all subaccounts of the current user.
<code>transfer_credit(recipient: str, amount: float)</code>	Transfer credit to another account.
<code>update_ssh_key(id: int, ssh_key: str)</code>	Update an SSH key.
<code>generate_pdf_invoices(quiet: bool = False, start_date: Optional[str] = None, end_date: Optional[str] = None, only_charges: bool = False, only_credits: bool = False)</code>	Generate PDF invoices based on filters.

For a complete list of methods and their usage, please refer to [Commands](#).



>

Powered by **mintlify**